

**Airspeed Limit [Kts]**

|                                  |             |      |       |
|----------------------------------|-------------|------|-------|
| Vne.....                         | 236         |      |       |
| Vno.....                         | 185         |      |       |
| Va.....                          | 159         | Duke | Calco |
| Vfe.....                         | 15° → 25°   | 153  | 162   |
| .....                            | 40°         | 130  | 130   |
| Vle.....                         | extend      | 128  | 153   |
| Vlo.....                         | retract     | 128  | 153   |
| .....                            | emrg extend | 128  | 153   |
| Vmca .....                       | 72          |      |       |
| Vxse.....                        | 104         |      |       |
| Vyse.....                        | 107         |      |       |
| Vsse.....                        | 92          |      |       |
| Wiper.....                       | 127         |      |       |
| Vx.....                          | 84          |      |       |
| Vy.....                          | 101         |      |       |
| Vr (normal, 0° flap).....        | 85          |      |       |
| Vr (soft / short, 15° flap)..... | 76          |      |       |

Maximum speed for boot inflation is 213 Kts CAS.

**AFM Operating Tips:** In extreme turbulence reduce power settings to obtain design maneuvering speed.

**Fuel Limit [U.S gal]**

AvGas Capacity W/O Nac tanks.....(Total) 192 (Usable) 182  
 AvGas Capacity W Nac tanks.....(Total) 246 (Usable) 236  
 Capacity **per wing** .....(OB Total) 40 (OB Usable) 38 | (Nac Total 27) (Nac Usable 27) | (IB Total) 56 (IB Usable) 53  
 Fuel Pressure.....**34 – 55 PSI**  
 Fuel transfer from Nac tanks 55 min

**RULE OF THUMB:** 1L / MIN / ENGINE @ CRUISE, for more accuracy do your math.

**NOTE:** The fuel should be balanced within 200 lbs. prior to departure.

**Oil Capacity [U.S quarts]**

12 (each engine) SOP 10 minimum (each engine) for dispatch  
 Oil Pressure 25 – **60 – 90** - 100 PSI  
 Oil Temperature 50 - **120 – 245 F**

**Purpose:** Lubricates – Cooling – Cleaning

**NOTE:** Make sure when reading on the oil Dipstick that you read the appropriate indication for the appropriate engine. Oil dipsticks are interchangeable between RH and LH Engines. On the same dipstick you will find right and left indication.

**Weight Limitation [lbs.]**

Max ramp weight.....7448  
 Max T/O weight.....7368  
 Max L/D weight.....7368 or Max L/D weight W/O BLR .....7000  
 Max weight Baggage compartment  
     Forward (Nose).....200 → 100 [lbs./sq ft]  
     Aft.....200 → 100 [lbs./sq ft]  
     Nacelle compartment.....150 (each) → 10 [lbs./sq ft]  
     Nacelle compartment.....50 (each) If NAC Tanks Full Fuel → 10 [lbs./sq ft]

**WARNING:** Nacelle compartments are made of fiberglass. **DO NOT EXCEED 10 [lbs./sq ft]**

**WARNING:** **DO NOT EXCEED 100 LBS. DIFF. BETWEEN R & L LOCKERS**

**Power Plant Limitation LTIO(R) / TIO(L)-540- J2B or J2BD**

Maximum T/O Power.....49 MP / 2575 RPM / Rich / Max HP 350 / Max CHT 500  
 Maximum MP [inches of Mercury] ..... To 15,000 49 - 15,000 to 22,000 49 minus 0,6/1000ft increase then minus 2,2 to 24,000  
 Maximum continuous Power....40 MP / 2400 RPM / Mixture rich / Max HP 315 / Max CHT 475  
 Maximum MP [inches of Mercury] ..... To 18,700ft 40 - 18,700 to 24,000 40 minus 1,7/1000ft increase  
 Max EGT.....1650 F  
 External Power (GPU) .....28 VDC with Master OFF

The propeller will tend to move toward feather position as soon as engine oil is lost. **Put to feather > 1000 RPM**

**Power Setting**

T/O Power setting.....49 MP/ **2575 RPM** / Mixture Rich  
 Climb Power setting.....38 MP/ **2400 RPM** / Mixture Rich  
 Cruise Power setting.....28 MP/ **2200 RPM** / Mixture Rich of Peak ~ (-50°C)

**Power setting Practical Tips:**

- Cruise power Technique.....Power ~ (26/27) MP → **2200 RPM** → Power will increase around ~28 MP → **Rich Peak ~ (-50°)** → Adjust
- Make sure that propellers are synchronized, when Props are not synchronized Navajo is very noisy and uncomfortable for customers ears.

**BLR Kit**

4 engine nacelle strake / 88 Vortex generators on the control surfaces / modification to the MLG

- **Result :** Improve stall characteristics by reducing (Vs,Vmca) speeds and increase (T/O,L/D) weights
- **Missing Vortex:** If Less than 84 vortex generators are in place, the aircraft must be operated in accordance of **original AFM limitations**.

**PA31-350 Chieftain / BLR modified / Limitations / Systems / Practical Tips**

**NOTE:** THIS DOCUMENT **DOES NOT** REPLACE THE AFM AND **SHOULD BE** USED AS A STUDY REFERENCE ONLY.

**NOTE:** LIMITATIONS / CONFIGURATIONS COULD BE DIFFERENT FROM ONE AIRPLANE TO ANOTHER

PLS CONFIRM WITH: AFM / FLIGHT SUPPLEMENTS / PLACARDS IN THE AIRPLANE, IF ANY DOUBTS

**BLR:** Boundary Layer Research. Quite simply, the name of the company.

**GPU:** Ground Power Unit

**Nayak:** Brand Name.



## Engines and Accessories

L (CW) - R (CCW).viewed from pilot seat, **No critical engine**

- Geared Starter (with starter ring)
- Single Driven dual Magnetos (some with 2 magnetos Ea. side)
- 28-volt 70-amp Alternator
- Shielded ignition
- Turbochargers
- Hydraulic pumps
- Oil filters
- Oil coolers
- Pneumatic pressure pumps
- 3 bladed propellers
- **Density controller:** Produce full rated power up to critical altitude (49 inHg. / 350 Hp.)
- **Differential pressure controller:** Automatically adjust MP during climb and descent, Prevent the need for throttle adjustment by the pilot

**Turbocharger compressor failure** → Revert to normally aspirated air, under this conditions 75 % of normal rated power or 262 HP will be available at sea level.

## Propellers

- 3 blades : Constant speed Controllable pitch and full feathering (**above 1000 rpm**)
- Propeller governor

**High Pitch ( decreased rpm & feathered position ) :** Nitrogen or air pressure, spring and blade counter weights

**Low pitch (increased rpm) :** Opposing governor regulated oil pressure

**NOTE:** Make all (Throttle / Prop) changes slowly and smoothly.

**NOTE:** Propeller will tend move to the feather position as soon as engine oil is lost.

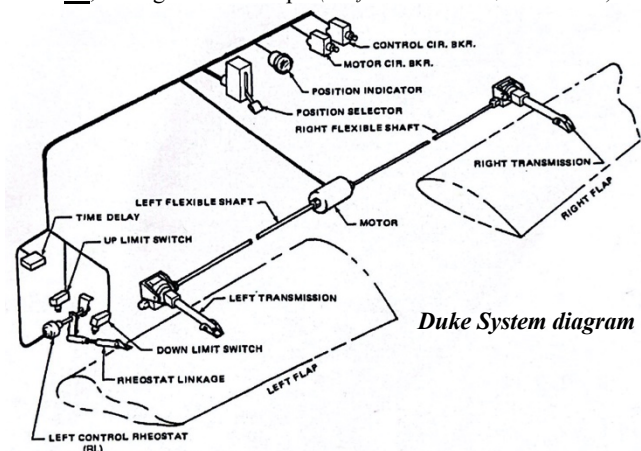
**CAUTION:** During an rpm overspeed condition, the engine should not be shut down since the propeller probably will not be featherable, and a high drag would result from a windmilling propeller in low pitch (Coarse).

**CAUTION:** Do not secure the engine if the propeller cannot be feathered.

## Flaps

Duke:

- 0 – 40°
- **Time delay** : relay stops the motor if asymmetric flap condition occurs, when flap do not extend together within 1 second ( approximately 9° right flap travel) the time delay relay will turn the flap motor off and the selector switch will become inoperative. The only way that further right flap extension can occur after the time delay switch shuts off the motor, is if the pilot would cycle the master switch off and on, and again select flaps. (Confirmed with Brad, maintenance)

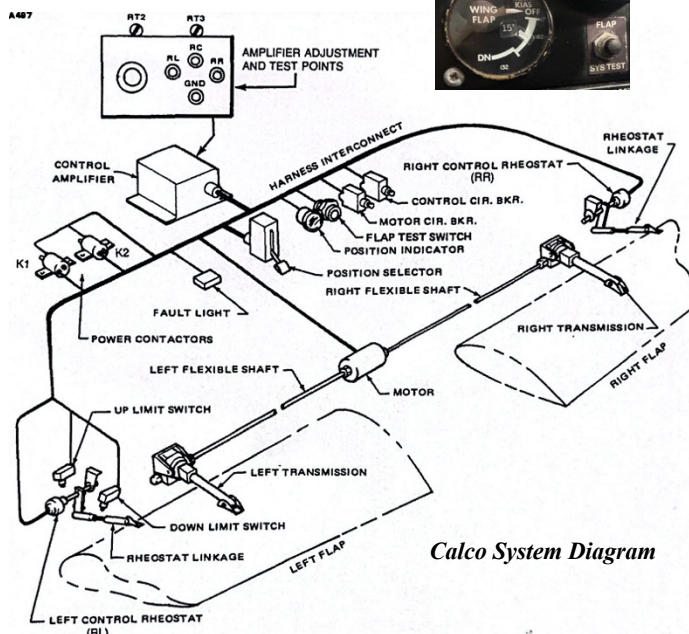


Duke System diagram

**CAUTION:** Do not reset time delay relay until A/C has landed and problem resolved as resetting the system and activation of the flap selector will allow another one second of operation allowing the asymmetric flap differential to become more severe.

Calco:

- 0 – 40° with 2° increments
- Right flap rheostat must agree with the left rheostat within 5° at all time or the system will shut down and cannot be re-activated until the cause has been corrected.
- The system also incorporates an **annunciator lights : FLAP**
- The entire system is electronically monitored and will prevent asymmetric flap position greater than 5°
- Flap Fault Test Switch (Check Figure 3): Test the integrity of the system not just the **annunciator Bulb**
  - Light = Pass = system working good
  - No light = Bad = Fault in the system
- The activation of this switch while flaps are in motion will give a false signal to the right side follower potentiometer, simulating an out-of-sync condition causing the amplifier to shut the system off and illuminate the fault light. Release of the switch will clear the simulated fault and allow the system to respond normally to selector position command. (PIPER NAVAJO SERVICE MANUAL)



Calco System Diagram

- Calco / Duke Flap **asymmetry** flying considerations:

**WARNING:** Flight with maximum flap asymmetry (0° and 25°) will require an aileron deflection of approximately 85% of the total available travel to maintain wings level. Considering the decreased lateral control maneuvering capability (in the direction of the maximum deflected flap), minimize bank angles and roll rate while maneuvering.

**WARNING:** Although flight tests have shown the aircraft to be controllable with symmetrical power, brief application of asymmetric power (increased power on the side of lesser flap deflection) may enhance roll control. Prolonged flight with large power asymmetry is not recommended.

**WARNING:** Considering the aircraft's reduced roll control capability, always select a landing runway which will minimize the crosswind component. Should a landing become necessary in a crosswind condition, maintain a wings level attitude while varying the aircrafts heading for runway alignment until the aircraft has been touched down.

**NOTE:** The landing approach should be planned to provide a higher and wider pattern than normal considering the aircraft's reduced

*lateral control capability. After turning final, configure the aircraft for landing. Set the power for a 500-800 ft. rate of descent and establish a final approach airspeed of 100 KTS. During the landing flare, gradually reduce power. Touchdown in a level flight attitude.*

**NOTE:** As the autopilot may mask high control forces necessary to counter an asym-metric ("split") flap condition, actuate the flaps only while the airplane is under manual control - not while the autopilot is engaged. Autopilot operation not authorized with greater than 15° of flap extension.

### **Landing Gear**

- Hydraulically actuated, air-oil oleo struts
- Total turning Arc 40° (20° either direction)
- Nose gear, shimmy damping device as gear retract
- **Red light :**
  - Transition between up lock and down lock position
  - Inboard gear door remains open when the gear is retracted
- **Green lights :**
  - Each individual gear is down and locked.
- **No Indication:**
  - Up and locked
- Gear extension or retraction will occur **normally in 6 seconds.**
- Each indicator lights can be dimmed individually by turning each light
- Gear **unsafe horn** will sound if :
  - MP < 12 MP and,
  - Gear retracted or not down and locked, or
  - Gear selector is up or up neutral when airplane on the ground and master ON

**IMPORTANT:** Never Ever Tow the airplane without removing the pin.

**NOTE:** Avoid unnecessary sharp turns, as excessive tire wear will result.

**Practical Tips:** When the desired gear position is obtained, the selector handle is forced back to the center / neutral position with the hydraulic force. The selector neutral position is an indication that the desired cycle is completed, and missing a light could be probably be a burnt out light bulb. But Landing gear position lights should be used as primary indications

### **Brake System**

- Hydraulically actuated by individual master cylinders mounted on the left set rudder pedal. (Optional on the right)
- A hydraulic reservoir separate from the main hydraulic system supplies fluid to each master cylinder.

**WARNING:** Braking may not occur if parking brake handle is pulled prior to brake pedal application

### **Hydraulic System**

- 2 hydraulic pumps (engine driven) : supplies pressure for gear operation, supplied through a reservoir
- Power pack : Central control unit
- Emergency hand pump
- Actuating cylinders
- Filters

**Electrical failure :** The doors will go to the open position with the hand pump and gear can be lowered.

### **Fuel System**

- Fuel cells (2 Main, 2 Outb., 2 optional 'Nayak' nacelle tanks)
- Engine driven fuel pump
- Emergency fuel pump (T/O-L/D-O/S-Prime or Fuel pressure < 34 PSI )
- Low pressure fuel boost pump

- Control valve
- Filters
- Fuel Pressure Gauge
- Fuel Flow Gauge
- Fuel drains ( 1/Tank + 1 Xfeed + 2 filter bowls), **total 7 or 9**
- Non-icing fuel tank vents

**Fuel Path :** Fuel cells → selector valve → Fuel Filter → Low Fuel boost pump → Emergency fuel pump → Fire wall shutoff valve → Engine driven Fuel pump → Injectors → Cylinders.

**Low Fuel Boost Pump :** Continuously operated and are provided to maintain fuel under pressure to other fuel pumps, improving the altitude performance of the fuel system, controlled by a CB. **Warning light illuminates** if fuel boost pump P < 3 PSI. If **Warning light** ON Use Emergency fuel Pump then can be turned OFF if Engine Fuel P > 34 PSI.

### **Nacelle Tank, Nayak, Limitation :**

- **Do not** transfer fuel until main tanks are at **least one-half full** or less.
- Approximately **55 minutes** are required to transfer all the fuel out of the nacelle tanks. (Full Fuel Tank)
- **Inside Nacelle locker door :** Baggage **limited to 50 pounds** capacity when nacelle tanks carry **FULL FUEL.**

**NOTE:** You cannot T/O with inop engine driven fuel pump or inop emergency fuel pump.

### **NOTE: FUEL MANAGEMENT**

- **Main tanks must be used for takeoff and landing.**
- **If the airplane is loaded with a rearward CG outboard tanks should be used first.**
- **This procedure will tend to move the CG forward with fuel burn-off.**

**NOTE:** The crossfeed system is not to be used for normal operation. When the crossfeed valve is on, be certain fuel selector valve on tank not in use is off. Do not use crossfeed to compensate for an inoperative emergency fuel pump.

**NOTE:** Do not climb or descent, fuel selectors on Outboard position. Prolonged slips or skids of 30 seconds or more, in any pitch attitude or other unusual or abrupt maneuvers which could cause uncovering of the fuel outlet must be avoided when outboard tanks are being used or when inboard tanks are less than one-quarter full.

### **WARNING:**

- **If L or R low boost (low fuel flow) warning light illuminates and Fuel gauge indicates fuel remaining in the corresponding inboard Tank → Malfunction of the flapper door in the inboard Tank → Immediately switch the outboard tank or select crossfeed to avoid FF interruption.**
- **The warning light will be on for a minimum of 10 seconds and will remain on if the cause is not corrected.**

### **Electrical System**

- Alternator
- Starters (**Max 15 (Steve) seconds of continuous cranking) but in the AFM 30 seconds, Follow Steve ☺.**)
- Ammeter
- Battery
- Voltage regulators (1/Alternator)
- External power source receptacle
- 28V DC System

**NOTE:** If a breaker pops it is recommended to allow the breaker to cool for a couple of minutes (two to five minutes) before resetting.

**NOTE:** Starter manufacturers recommend that cranking periods be limited to 30 seconds with a two minute rest between cranking periods. Longer cranking periods will shorten the life of the starter.



- **Primary electrical source**

**Alternators :** Two 28V - 70 amp alternators. Controlled independently by two voltage regulators. *(Confirmed with Brad, maintenance)*

**CAUTION:** The alternator output circuit breakers or the voltage regulator CB **should not** be opened manually when the alternators are functioning properly. *(Confirmed with Brad, maintenance)*

**NOTE:** When flying in heavy rain, the electrical load on the **right alternator must be reduced to 40-amperes or less** to ensure against alternator belt slippage.

**NOTE:** Alternator output will not be checked until after external power disconnect

**Ammeter :** Indicates both battery charging current and alternators output.

Needle left of center → battery discharging.

Needle right of center → battery charging.

During single engine operation the feature can be used to determine how much the electrical load should be reduced. To check the output of each alternator use the **press-to test button** located on either side of the ammeter.

- **Secondary electrical source**

**Batteries :** Standard 24V - 20 amp *(Confirmed with Brad, maintenance)*

- **External Power source**

28V DC, **Turn OFF** the Master switch before inserting the external power. MAX 30V DC ( SOP ) *(Confirmed with Brad, maintenance)*

**NOTE:** If the aircraft battery is weak, charging current will be high. Do not takeoff until charging current falls below 20 amperes. Do not takeoff with a completely discharged battery as three volts are needed to excite the alternator.

- **Landing and taxi lights**

During gear retraction, if the lights have not been turned OFF manually they will be turned OFF automatically because they are **wired through the nose gear down micro switch**. If the switch is left ON, when the gear is extended the lights will again illuminate. *(Confirmed with Brad, maintenance)*

**Instrument Panel**

**Standard attitude indicator & Heading indicator :** operated by air ( Pneumatic ).

**Turn and slip indicator :** operated electrically to serve as stand by instrument in case the pneumatic system becomes inoperative

**Annunciator Panel**

**Press-to-test :** 3 second → Delay for low fuel flow warning and will remain 10 after the others have extinguished.

**Flap fault test switch (Calco only):** independent and located near the flap gauge



**Pneumatic System**

- 2 Engine driven pneumatic pump, supplies:
  - Driven Gyros ( attitude / heading indicator)

- Deicing boots

- 5.5 ± 0.4 in of Hg @ 2200 RPM

**Wings and empennage deicing**

- **Equipment:**

- Heated pitot tubes (Dual)
- Electrically heated windshield (Captain side only)
- Electric propeller anti-icing strip
- Deicing boots on wing a empennage (H/V Stabilizer) leading edges
- Wing ice light (Left wing only)
- Constant suction is applied to all of the surface deicer boots to provide smooth streamline leading edge with deicing system OFF.
- Maximum speed for boot inflation is 245 MPH (213 Kts) CAS.
- Wing De-ice switch ON → one complete inflation cycle, the boots will be inflated for approximately 6 seconds or 12 seconds *(Confirmed with Brad, maintenance)*
- A blue indicator light with press-to-test and dimming feature illuminate when the boots inflate to a predetermined pressure 18 PSI.
- When the de-icing system is activated no CO-Pilot Gyros.
- - 20° C Using or testing the Deicing boots is not recommended. Can crack the rubber and won't pick up any ice in that low temp.
- Pitot tube heat **should not be** used for extended periods during ground operations
- The electrically heated windshield **should not be** operated for extended periods during ground operation unless windshield ice is present to prevent overheating.

**Heat, Ventilating and Defrosting system**

*The section below is confirmed with Brad, maintenance.*

- **Standard Basic configuration:**

- FWD R/H combustion Heater (Janitrol)
  - Janitrol heater installed in the right nose section furnishes Hot air for cabin heating windshield defrosting.
  - Fuel is supplied from the R/H I/B wing tank, **2 gal/h.**
- Recirculating air system
- Ground vent system

- **Optional configuration:**

- FWD L/H (Airborne heater)
  - Has a dedicated fuel pump in the L/H wing root drawing fuel from the L/H I/B wing tank **2 gal/h.**
  - A/C must have weight off wheels and intake airflow must be provided for heater to function
  - Has cockpit airflow cable and On/Off switch for controls on L/H instrument panel
- AFT combustion heater (Janitrol)
  - Has dedicated fuel pump in R/H wing root drawing fuel from the R/H I/B wing tank
- L/H & R/H exhaust heater (Muff heater):
  - All SkyCare's planes have L/H engine heater muff systems, and some have R/H muffs
- Air conditioning :
  - The air scoop opens on the ground with the air conditioning system activated, and closes automatically once the R/H gear

safety switch is contacted and weight is off wheels

**CAUTION:** The heater air inlet valve must be full open prior to and during heater operation. If this valve is not full open during heater operation the heater will become inoperative from overheating.

**CAUTION:** If the heater is turned off after operation on the ground, the switch should first be turned to the "FAN" position for 3 minutes to cool (SOP) the heater, then turned to off. If the heater is turned off in the air it is not necessary to select the "FAN" position.

#### Stall warning

Stall warning is given at about 4 to 10 Kts before an actual stall would occur.

On some airplanes the stall detector vane can be heated when Left Pitot heat switch is turned ON. But only in flight.

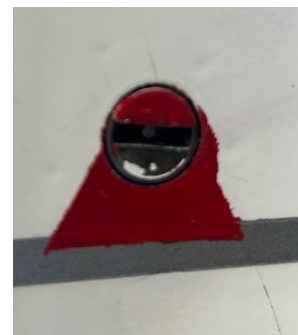
- An electrically operated windshield wiper (pilot's side only) is available with the heated windshield installation. The windshield wiper should not be operated at speeds above 127 Kts or on a dry surface.

#### Flight Preparation

##### Practical tips:

- Check log book for snags / Deferred Items / time remaining / check the last 5 flight pages for snags
- Walk around / Run-up
  - If any snags just got fixed, **focus on the fixed snag** during run up
- Take water - commissary - Lunches

**CAUTION:** Check that the four screws securing the nose bowl are present and secure. A screw driver may be needed to do this. Check that the six (three per side) side cowl fasteners are present and secure. Check for both the proper alignment of the paint stripes red on red on the cowling and fastener and for the extension of the locking pin indicator in the fastener slot.



Go ! Engine cowl secured

No Go!! Engine cowl NOT secured

**WARNING:** Failure to ensure the presence and security of nose bowl and cowl side fasteners may result in inflight cowl separation and loss of aircraft control.

#### Loading

- **Light load :** Keep the weight aft
- **Medium Load :** Center the weight
- **Full load :** Concentrate as much forward as possible
- **Place Small Dense** pieces in the nose and wing locker

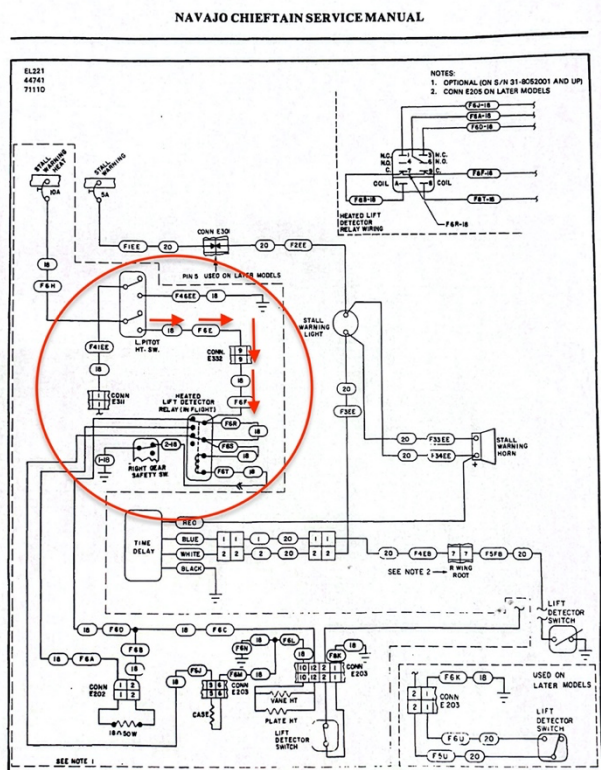


Figure 11-68. Stall Warning (With Time Delay)  
(S/N PA-31-7652077, and 31-7652079 to 8152113)

Revised: 2/8/82

3G4

ELECTRICAL SYSTEM

#### Electric Propeller Deicer

- Propeller deicer pads
- Modified starter ring gears
- Brush assemblies
- Timer
- Ammeter ( when switch ON → 8 to 12 amp )

- Outboard halves R.E : 30 sec
- Inboard halves R.E : 30 sec
- Outboard halves L.E : 30 sec
- Inboard halves L.E : 30 sec

Example: c, d, a, b, OR d,a,b,c

#### Anti-Static Wicks

- Anti-static wicks are installed on the trailing edges of the wing panels and tail surfaces. They are designed for efficiency in clearing the airplane of surface static electricity that might disrupt LF reception, ADF indication or cause VHF interference.

#### Windshield wipers